



CSA1708 – ARTIFICIAL INTELLIGENCE TECHNICAL PRESENTATION

REACTIVE AGENT-BASED TRAFFIC CONTROL SYSTEM

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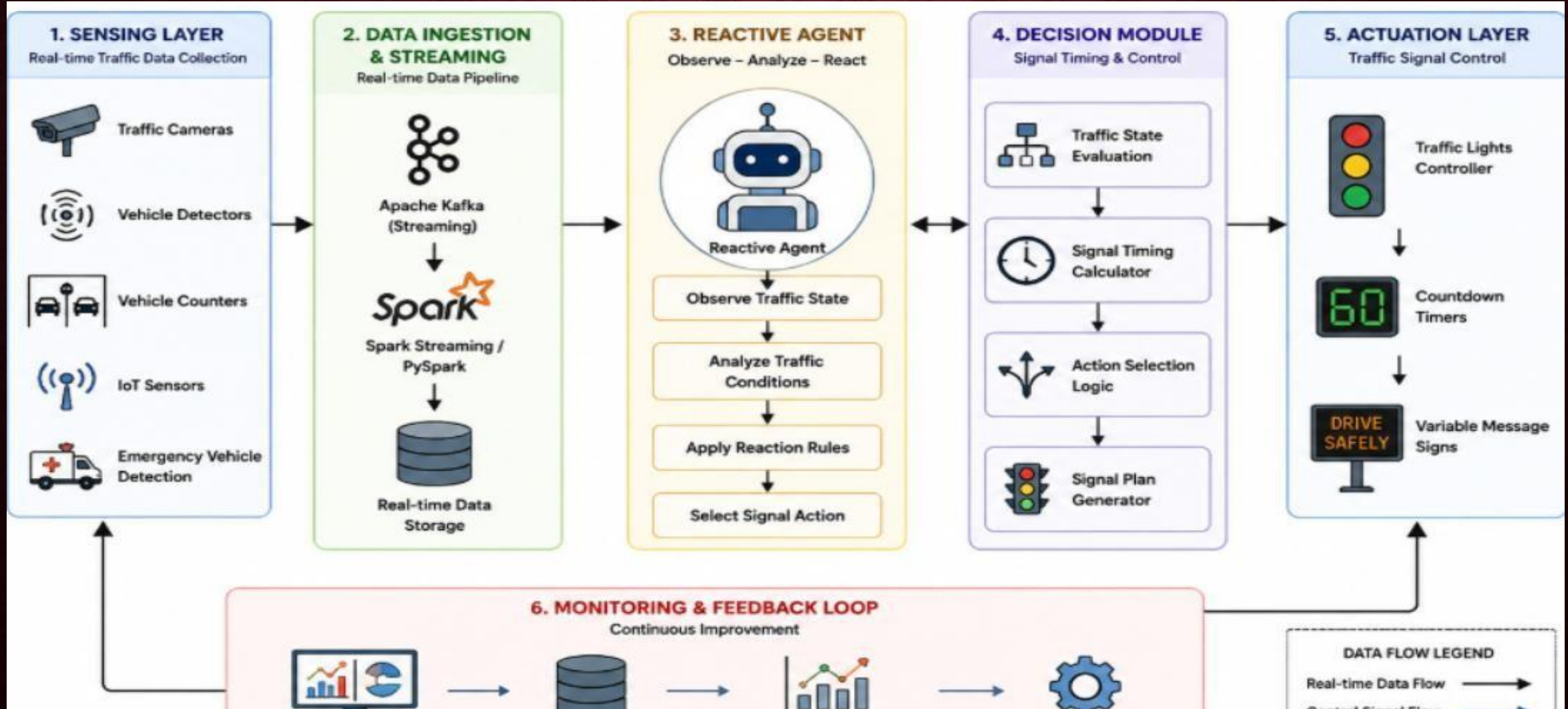
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INTRODUCTION

- A Reactive Agent is an intelligent system that responds immediately to changes in its environment.
- In smart traffic control, the agent continuously monitors real-time traffic conditions.
- Traffic inputs include vehicle density, waiting time, lane occupancy, and emergency vehicle presence.
- The agent analyzes current conditions and selects appropriate traffic signal actions.
- The main objective is to reduce congestion, waiting time, and unnecessary traffic delays.

SYSTEM ARCHITECTURE





REACTIVE AGENT WORKING

- The agent continuously receives the latest traffic information from the environment.
- It compares current traffic conditions with predefined response rules.
- When traffic density increases, the agent can extend the green signal duration.
- When traffic decreases, the agent can reduce unnecessary green time and adjust signal priority.
- The agent repeatedly observes and reacts without requiring a long-term prediction model.



APPLICATIONS



- **Smart Traffic Signals** – Dynamically adjusts signal timings based on real-time vehicle density.
- **Emergency Vehicle Priority** – Detects ambulances and fire vehicles and provides priority through signal control.
- **Highway Traffic Management** – Monitors traffic flow and reacts to congestion on major roads.
- **Intelligent Intersection Control** – Coordinates multiple traffic signals to improve vehicle movement at intersections.
- **Smart City Transportation** – Integrates real-time traffic data to support efficient and



CONCLUSION

A Reactive Agent-Based Traffic Control System continuously monitors real-time traffic conditions and responds immediately to changing vehicle flow. By processing streaming data from sensors and cameras, the agent dynamically adjusts traffic signals to reduce congestion and waiting time. It can also provide priority to emergency vehicles and improve traffic movement at intersections. The integration of AI agents, IoT sensors, and real-time data processing enables efficient and intelligent traffic management.

THANKYOU